

I believe that the preceding discussion has brought us so closely to this definition that it needs no additional explanation. I would only like to emphasize that the concept of the continuity of a function is essentially local. Similarly to the concept of the limit of function, it is related to a particular point x . A function may be either continuous at all points of an interval over which it is defined, or discontinuous at some of its points.

Taking the examples given above, can you single out those functions that are discontinuous at particular points?

Reader: To begin with, I may refer to the function whose graph is plotted in Fig. 28. This function is discontinuous at $x=0$.

Author: Why?

Reader: Because at this point the function assumes the value $y=1$, though the limit of this function at this point is apparently zero.

Author: Very good. Can you give other examples?

Reader: The function $y=\{x\}$ (see Fig. 21) is discontinuous at points $x=0, \pm 1, \pm 2, \pm 3, \dots$. The function $y=\sin \frac{1}{x}$ (see Fig. 24) is discontinuous at $x=0$ where it is undefined and, moreover, has no limit. The function whose graph is shown in Fig. 14a (see the previous dialogue) is discontinuous at $x=2$. The function $y=\tan x$ is discontinuous at points $x=\pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \pm \frac{7\pi}{2}, \dots$.

Author: That will do. Note that the points at which the continuity of a function is violated are called discontinuity points. We say that at these points a function has a discontinuity. In passing through a discontinuity point a graph of a function manifests a singularity. This fact is well illustrated by the examples you have just indicated.

Reader: The discontinuity points in all these examples result in an interruption of the curve plotting the function. One exception is the function $y=\sin \frac{1}{x}$ since it is simply impossible to trace a graph of the function at $x=0$.

Author: I may add that neither could you plot the function $y=\tan x$ at its discontinuity points (since you cannot draw a line which "goes into infinity").